$\Rightarrow \quad x^2 + \frac{1}{3}x - \frac{1}{3} = 0 \quad / \cdot 3$

$$\mathbf{3x^2 + x - 10 = 0}$$

2.7.5 : Sestavte kvadratickou rovnici o kořenech $x_1 = \frac{1}{2};\ x_2 = -\frac{2}{5}$

$$\begin{aligned} V.v. \Rightarrow \quad x_1 + x_2 &= -b \quad \wedge \quad x_1 \cdot x_2 = c \\ \frac{1}{2} - \frac{2}{5} &= -b \qquad \frac{1}{2} \cdot \left(-\frac{2}{5}\right) = c \\ b &= -\frac{1}{10} \qquad\quad c = -\frac{1}{5} \end{aligned}$$

$$\Rightarrow \quad x^2 - \frac{1}{10}x - \frac{1}{5} = 0 \quad / \cdot 10$$

$$\mathbf{10x^2 - x - 2 = 0}$$

2.7.6 : Sestavte kvadratickou rovnici o kořenech $x_1 = 1,5;\ x_2 = 3,5$

$$\begin{aligned} V.v. \Rightarrow \quad x_1 + x_2 &= -b \wedge x_1 \cdot x_2 = c \\ \frac{3}{2} + \frac{7}{2} &= -b \qquad \frac{3}{2} \cdot \frac{7}{2} = c \\ b &= -5 \qquad\quad c = \frac{21}{4} \end{aligned}$$

$$\Rightarrow \quad x^2 - 5x + \frac{21}{4} = 0 \quad / \cdot 4$$

$$\mathbf{4x^2 - 20x + 21 = 0}$$

2.7.7 : Sestavte kvadratickou rovnici o kořenech $x_1 = 6 + \sqrt{2};\ x_2 = 6 - \sqrt{2}$

$$\begin{aligned} V.v. \Rightarrow \quad x_1 + x_2 &= -b \quad \wedge \quad x_1 \cdot x_2 = c \\ 6 + \sqrt{2} + 6 - \sqrt{2} &= -b \qquad (6 + \sqrt{2})(6 - \sqrt{2}) = c \\ b &= -12 \qquad\qquad 36 - 2 = c \\ &\qquad\qquad\quad c = 34 \end{aligned}$$

$$\Rightarrow \quad \mathbf{x^2 - 12x + 34 = 0}$$

2.7.8 : Sestavte kvadratickou rovnici o kořenech $x_1 = -3\sqrt{3};\ x_2 = 2\sqrt{3}$

$$\begin{aligned} V.v. \Rightarrow \quad x_1 + x_2 &= -b \wedge \quad x_1 \cdot x_2 = c \\ -3\sqrt{3} + 2\sqrt{3} &= -b \qquad (-3\sqrt{3}) \cdot 2\sqrt{3} = c \\ b &= \sqrt{3} \qquad\quad c = -18 \end{aligned}$$

$$\Rightarrow \quad \mathbf{x^2 + \sqrt{3}x - 18 = 0}$$

2.7.9 : Sestavte kvadratickou rovnici, která má kořeny trojnásobné než jsou kořeny rovnice $x^2 - 3x + 2 = 0$, aniž danou rovnici řešíte.

$$V.v. \Rightarrow \begin{array}{l} X_1 + X_2 = -b \\ \underline{X_1 \cdot X_2 = c} \end{array} \leftarrow \begin{cases} X_1 = 3x_1 \\ X_2 = 3x_2 \end{cases}$$

$$\Rightarrow \begin{array}{l} 3x_1 + 3x_2 = -b \\ \underline{3x_1 \cdot 3x_2 = c} \end{array}$$

$$\begin{array}{l} 3(x_1 + x_2) = -b \\ \underline{9x_1 \cdot x_2 = c} \end{array} \leftarrow \begin{cases} V.v. \Rightarrow x_1 + x_2 = 3 \\ \qquad\quad x_1 \cdot x_2 = 2 \end{cases}$$

$\Rightarrow \quad 3 \cdot 3 = -b \quad \Rightarrow \quad b = -9$

$9 \cdot 2 = c \quad \Rightarrow \quad c = 18$

$$\mathbf{x^2 - 9x + 18 = 0}$$

2.7.10 : Sestavte kvadratickou rovnici, která má kořeny čtyřikrát větší než jsou kořeny rovnice $x^2 - 9x + 15 = 0$, aniž danou rovnici řešíte.

$$V.v. \Rightarrow \begin{array}{l} X_1 + X_2 = -b \\ \underline{X_1 \cdot X_2 = c} \end{array} \leftarrow \begin{cases} X_1 = 4x_1 \\ X_2 = 4x_2 \end{cases}$$

$$\Rightarrow \begin{array}{l} 4x_1 + 4x_2 = -b \\ \underline{4x_1 \cdot 4x_2 = c} \end{array}$$

$$\begin{array}{l} 4(x_1 + x_2) = -b \\ \underline{16x_1 \cdot x_2 = c} \end{array} \leftarrow \begin{cases} V.v. \Rightarrow x_1 + x_2 = 9 \\ \qquad\quad x_1 \cdot x_2 = 15 \end{cases}$$

$$\begin{aligned} \Rightarrow \quad 4 \cdot 9 &= -b \quad \Rightarrow b = -36 \\ 16 \cdot 15 &= c \quad \Rightarrow c = 240 \end{aligned}$$

$$\mathbf{x^2 - 36x + 240 = 0}$$

2.7.11 : Sestavte kvadratickou rovnici, která má kořeny opačné než jsou kořeny rovnice $x^2 - x - 6 = 0$, aniž danou rovnici řešíte.

$$V.v. \Rightarrow \begin{array}{l} X_1 + X_2 = -b \\ \underline{X_1 \cdot X_2 = c} \end{array} \leftarrow \begin{cases} X_1 = -x_1 \\ X_2 = -x_2 \end{cases}$$

$$\Rightarrow \begin{array}{l} -x_1 - x_2 = -b \\ \underline{(-x_1) \cdot (-x_2) = c} \end{array}$$

$$\begin{array}{l} x_1 + x_2 = b \\ \underline{x_1 \cdot x_2 = c} \end{array} \leftarrow \begin{cases} V.v. \Rightarrow x_1 + x_2 = 1 \\ \qquad\quad x_1 \cdot x_2 = -6 \end{cases}$$

$$\begin{aligned} \Rightarrow \quad 1 &= b \\ -6 &= c \end{aligned}$$

$$\mathbf{x^2 + x - 6 = 0}$$

2.7.12 : Sestavte kvadratickou rovnici, která má kořeny opačné než jsou kořeny rovnice $4x^2 - 11x + 5 = 0$, aniž danou rovnici řešíte.

$$\begin{aligned} 4x^2 - 11x + 5 &= 0 \quad / : 4 \\ x^2 - \frac{11}{4}x + \frac{5}{4} &= 0 \end{aligned}$$

$$V.v. \Rightarrow \begin{array}{l} X_1 + X_2 = -b \\ \underline{X_1 \cdot X_2 = c} \end{array} \leftarrow \begin{cases} X_1 = -x_1 \\ X_2 = -x_2 \end{cases}$$

$$\Rightarrow \begin{array}{l} -x_1 - x_2 = -b \\ \underline{(-x_1) \cdot (-x_2) = c} \end{array}$$